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CLINICAL UPDATES

Management of paracetamol (acetaminophen) toxicity

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What you need to know

- Paracetamol overdose is time critical: Acetylcysteine is almost universally effective when given within 8-10 hours of ingestion (with an estimated overall case fatality rate of 0.4%), while delay substantially worsens outcomes
- Replace the traditional 3-bag treatment regimen with modern 2-bag regimens (eg, SNAP or Australia/New Zealand) to reduce treatment delays and adverse reactions
- Consider using sodium chloride as the acetylcysteine diluent and/or reduced volume infusions in adults and children with body mass <40-50 kg to prevent hyponatraemia
- Do not stop treatment for mild anaphylactoid reactions. Pause, administer antihistamines, and restart acetylcysteine at a slower rate
- Do not delay mental health assessment for "medical clearance"

Case vignette

You see a 22 year old woman three hours after her arrival at the emergency department. She is experiencing psychological distress and reports that, nine hours earlier, she swallowed "handfuls" of paracetamol tablets over a two hour period, intending to end her life. This is not her first suicide attempt: when she previously intentionally overdosed on paracetamol, she experienced an unpleasant anaphylactoid reaction with the antidote acetylcysteine and is therefore reluctant to start treatment. Her blood test results show a paracetamol concentration much higher than your nomogram's treatment line. You recognise the large size of the overdose, prolonged ingestion duration, need for treatment, and clinical dilemma posed by her prior adverse reaction. You consider your next steps.

Globally, paracetamol (acetaminophen) overdose has an estimated incidence of 7.4 per 100 000 population per year, though this varies by region; incidence in the UK may be as high as 156.7 per 100 000.[1,2] Longitudinal data suggest these rates may be increasing.[3] Poisoning is most common in women and girls, and incidence peaks in the age bracket 20-29, although men, boys, and older people often take larger overdoses.[2,4]

Most patients recover fully with timely acetylcysteine treatment. Based on pooled proportions from international observational datasets, mortality following treated overdose is estimated at 0.4% (95% confidence interval 0.1% to 2.3%) globally, with acetylcysteine almost universally effective when administered within 8-10 hours of ingestion.[1] Even when acetylcysteine is administered within that time window, a small proportion develop acute liver failure, which carries a high mortality rate and remains a leading indication for hyperacute liver transplantation.[1,4-6] This article describes recent developments in evidence, guidelines, treatment regimens, and management strategies to support readers in making informed decisions regarding paracetamol overdose.

Clinical presentations of paracetamol toxicity

Most patients present within 24 hours of ingestion.[4] In the 24 hours following overdose, most are asymptomatic or may experience mild symptoms including nausea and vomiting.[7] Both patients and clinicians can underestimate the severity of the situation, as patients may appear physiologically stable despite evolving hepatocellular injury. As acute liver injury progresses to 24-72 hours post-ingestion, patients may develop right upper quadrant pain and hepatic tenderness, while jaundice is rare [8]. Acute liver failure may develop after several days, with risk primarily driven by ingestion dose and time to antidote. While disaggregated prevalence data on acute liver failure in the setting of paracetamol overdose are lacking, observational data from the US suggest 39% of acute liver failure cases are the result of paracetamol overdose.[9] Registrations for acute liver transplantation in patients with paracetamol overdose are approximately 0.016 per 100 000 population per year.[10]

Although the pattern above represents the typical clinical course, less common phenotypic presentations can be more challenging to recognise:

- Massive overdose—In the context of paracetamol toxicity, massive overdose is generally defined as ingestion of >30 g or a paracetamol concentration >300 mg/L. In very large ingestions, patients may present early with altered mental status or metabolic acidosis.[11] Vomiting may occur, likely related to the gastric irritant effects of phenol chemicals such as paracetamol
- Staggered overdose—Approximately one in three ingestions occur over sustained time periods (more than one to two hours), usually reflecting prolonged self-harm ingestions or supratherapeutic dosing.[4] These presentations are often higher risk relative to the dose ingested. Standard nomograms are unreliable here, necessitating lower thresholds for initiating treatment while liver injury severity is established[12]
- Undisclosed ingestion—Patients may present days after ingestion with jaundice, coagulopathy, hepatic encephalopathy, or coma

- Acute kidney injury (AKI)—AKI occurs in 2-10% of patients.[13] Direct acute tubular necrosis (ATN) driven by local CYP450 activation within the kidney can occur within 48-72 hours of excess paracetamol consumption, even in the absence of severe liver injury, and may be the predominant presenting feature in a small subset of patients. AKI can also develop as a component of hepatorenal syndrome in people with established acute liver failure.

Who is at the greatest risk?

Pathophysiology of risk

When paracetamol is consumed in excess, normal metabolic pathways (glucuronidation and sulfation) are saturated. Toxic amounts of the metabolite N-acetyl-p-benzoquinone imine (NAPQI) are formed by CYP450 enzymes.[14] Normally, liver glutathione neutralises NAPQI, but in overdose, glutathione stores are exhausted and NAPQI binds to mitochondrial proteins, causing oxidative stress, mitochondrial permeability transition pore opening, and necrotic cell death. This necrosis is centrilobular because these hepatocytes have the highest CYP enzyme activity. Risk stratification therefore aims to identify patients in whom NAPQI formation is likely to exceed glutathione detoxification capacity, such that acetylcysteine is required to replenish glutathione and prevent liver injury. Co-ingestion of paracetamol with other medications is common, and patients may benefit from monitoring for co-toxidromes, or delayed absorption from the gastrointestinal tract (box 1).

Box 1: Management in special circumstances

Massive ingestions

For massive overdoses (>30 g or concentration >300 mg/L), some observational data suggest higher acetylcysteine doses may reduce liver toxicity. The Scottish and Newcastle antiemetic protocol (SNAP) regimen delivers a higher dose of acetylcysteine over the first 12 hours than traditional regimens. The ongoing HiSNAP trial is investigating whether doses higher than standard SNAP further reduce toxicity in this population.[15 16] Haemodialysis is sometimes considered in people with massive overdose with severe metabolic acidosis and paracetamol concentration >900 mg/L, though evidence of benefit is limited. Haemodialysis clears paracetamol and improves acidosis but also removes acetylcysteine, requiring dose doubling.[17] Plasmapheresis (plasma exchange) has been used experimentally in some centres for massive overdose but lacks evidence of survival benefit.[18]

Modified release or delayed absorption

Extended release preparations, or co-ingestion of anticholinergic or opioid medications (common in mixed overdose), may delay acetylcysteine absorption. If operating a "150 line" nomogram, measuring repeat paracetamol concentrations four to six hours after the first is recommended, as paracetamol concentrations may surpass the nomogram line. Repeat paracetamol concentration testing is not typically recommended with more conservative "100 line" nomograms. Clinicians may be unaware of bi-layer preparations of immediate and extended release paracetamol, which can be associated with unpredictable toxicity and biphasic peak concentrations.[19] Potential for harm led to withdrawal of these products in Europe.[20] Discuss all bi-layer preparation ingestions with a poisons centre.

Pregnancy

In overdose, paracetamol toxicity is associated with fetal loss (though propensity weighted data are lacking[21]), but not fetal abnormalities. Acetylcysteine is safe and should be dosed based on the patient's current (pregnant) body mass.[22]

Intravenous overdose

The nomogram is not validated for intravenous (IV) overdose. UK guidelines suggest treatment with acetylcysteine using the SNAP regimen for paracetamol dosages >60 mg/kg or 4 hour paracetamol levels above a 50 mg/L nomogram line.[23]

Extremes of body mass

Acetylcysteine dosing is based on body mass, typically capped at 100-110 kg. Reduced fluid volumes are helpful to prevent fluid overload in patients with low body mass.[20] The use of sodium chloride as a diluent is discussed below.

Paediatrics

Intentional ingestions of paracetamol are common in adolescents.[4] Registry data suggest that paediatric acute liver failure following acetaminophen overdose is rarely fatal.[24] Dose acetylcysteine by body mass, and consider reducing fluid volumes, using sodium chloride as a diluent, as mentioned above. Younger patients benefit from the same reductions in anaphylactoid reactions as adults with the SNAP protocol.[4]

Investigations

Generally, initial investigations for suspected overdose include:

- Serum paracetamol concentration plotted on a modified Rumack-Matthew nomogram to determine whether acetylcysteine is required
- Liver profile, including alanine aminotransferase (ALT), alkaline phosphatase (ALP), and bilirubin to assess for established hepatocellular necrosis
- International normalised ratio (INR) to assess hepatic synthetic function
- Urea, creatinine, and electrolytes to assess for AKI
- Full blood count to establish haematological baseline
- Venous blood gas (VBG) to assess for metabolic acidosis or elevated lactate, which act as early markers of mitochondrial dysfunction in massive overdose.

The serum paracetamol concentration plotted on a modified Rumack-Matthew nomogram is the primary risk stratification tool to determine the need for treatment in people with a single acute overdose. The remaining investigations (ALT, INR, creatinine, VBG) may impact initial treatment decisions, for example, by signalling the need for acetylcysteine due to evidence of established or evolving liver injury, or early signs of AKI or acute liver failure requiring specialist input. These investigations are also essential to assess evolving hepatocellular injury, synthetic function, and complications during and after treatment, and to inform decisions about continuing or stopping acetylcysteine.

The nomogram

For single acute ingestions, starting acetylcysteine usually relies on plotting the paracetamol concentration against the time since ingestion on a modified Rumack-Matthew nomogram (fig 1).[26] The nomogram was derived from observational data of serum paracetamol concentration and reported time since ingestion, estimating hepatotoxicity risk (and consequently, need for treatment) using threshold lines with 4 hour half lives. Different lines are used globally, reflecting different risk tolerances; the UK uses a 100 mg/L line, while Australia and the US use a 150 mg/L line.[11 12 27]

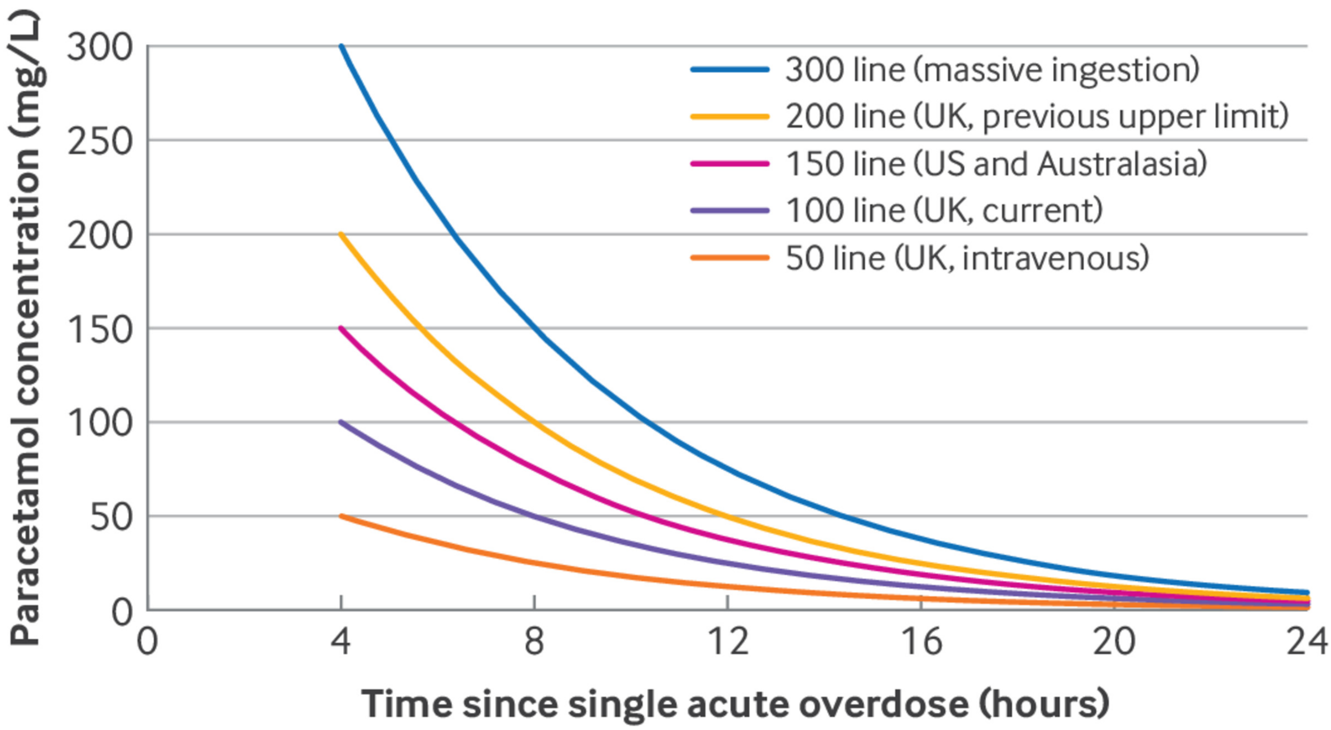

Fig 1 | Modified composite Rumack-Matthew nomogram, showing the variety of derived logarithmic lines in current clinical use internationally. All lines represent a 4 hour half life from original values. Interpretation of paracetamol concentration within the first four hours following ingestion is considered too unreliable for clinical use because of the risk of ongoing absorption. Values above the 300 line are typically considered very high risk for hepatotoxicity. The UK previously used a 200 line, with risk assessment applied when deciding on treatment between the 100 line and 200 line. The US Food and Drug Administration required a 25% reduction from the 200 line for an increased margin of safety, resulting in a 150 line which was also adopted in clinical use in some other countries.[25] The 100 line is used in the UK, with a 50 line playing a role in guidance regarding the assessment of risk following intravenous paracetamol overdose

Guidelines differ on what constitutes a single acute overdose that is suitable for plotting. UK and Australasian guidelines apply strict 1 or 2 hour ingestion windows, respectively, whereas recent US consensus guidelines include any ingestion period of less than 24 hours.[11 12 27] Follow specific local guidance and remain aware that staggered ingestions cannot be plotted on the nomogram; commence treatment for staggered ingestions immediately, with subsequent laboratory test results to guide discontinuation of therapy.

Nomogram alternatives are emerging to predict which patients treated with acetylcysteine will go on to develop significant liver injury (often without differentiating between types of ingestion), but lack sufficient prospective validation. For example, there is increasing interest in using presentation blood test results "paracetamol concentration × alanine aminotransferase" as a multiplication product for risk stratification, which may be useful for staggered ingestions and those that occurred more than 24 hours prior to presentation.[25] In cases of early presentation (<4 hours) following paracetamol overdose, wait until four hours have elapsed to draw initial paracetamol concentrations.

Limitations in resource constrained settings

In settings where plasma paracetamol quantification is not available, clinicians must rely on self-reported ingestion dose (treatment often initiated if >150-200 mg/kg), clinical symptoms, or raised liver transaminase levels.[28] While reliance on reported dose often leads to over- or underestimating actual ingestion dose, it is a pragmatic necessity.[29-31]

A major unmet need globally is the lack of early, predictive biomarkers that can accurately identify specific individuals from the large "at risk" group who are truly destined to develop severe liver injury. Current measures, including ALT and INR, rise only after damage has begun, often too late for ideal preventive treatment. Novel biomarkers such as keratin-18 and miR-122 show promise but are not yet in routine clinical use.[32]

Management of paracetamol toxicity

Treatment aims to reduce ongoing paracetamol absorption, enhance its elimination (in specific cases), and administer the antidote acetylcysteine.[33] Experimental treatments (eg, fomepizole use, as seen in the US) are described in box 2.

Box 2: Areas for further research

- *Prospectively predicting development of injury*—Developing novel biomarkers (eg, microRNAs, protein adducts, keratin-18) or novel predictive models to identify patients at high risk of clinically significant liver injury before ALT rises would support precision medicine approaches for the application and development of adjunct treatments that prevent the development of injury.[24 34]
- *Optimising antidote therapy*—Refining acetylcysteine dosing to support personalised approaches, robustly studying novel antidotes, and developing an evidence base examining emerging antidote practices (eg, use of double dose acetylcysteine or fomepizole) is a critical step in reducing severe liver injuries.[16 35 36]
- *Treating established injury*—The development of efficacious therapies that can reverse established liver damage and promote regeneration (eg, regenerative cell therapies) could reduce the requirement for orthotopic liver transplantation and its associated morbidity, and could potentially also reduce liver failure deaths in patients for whom transplantation is not an option.[37 38]

- *Predicting trajectory of established injury*—Finding prognostic biomarkers that predict who will spontaneously recover rather than follow a worsening trajectory would allow the targeted application of emerging regenerative liver therapies, and support clinical trial population enrichment.
- *Improving data quality*—Understanding the highest yield intervention points in care pathways requires resolving persistent deficits in routine emergency department data classification, patient tracking, and clinical context.[39 40] Developing harmonised, longitudinal data linkage frameworks across the care continuum is a critical prerequisite for translating policy into actionable operational improvements and validating future interventions.[41]
- *Primary and secondary prevention*—Novel research streams are being developed which may support the proactive identification of patients at risk for significant overdose and facilitate targeted preventive care. Data to support these practices requires significant novel co-operation across primary, urgent and emergency care systems.[42]
- *Knowledge translation*—Adoption of evidence based treatment changes into guidelines remains a slow, top down process.[43] Turning clinical studies into real world impact requires increasingly multimodal dissemination strategies to drive improvements in the healthcare experience of a cohort of patients for whom the prevention of further suffering should be a priority.[44] Recognition and management of paracetamol overdose as a time critical presentation remains poor.

Activated charcoal

A single dose of activated charcoal (50 g orally) administered within four hours of ingestion reduces paracetamol absorption: an observational study (ATOM-2; n=200) showed that activated charcoal reduced hepatotoxicity (development of ALT>1000 U/L) when given to patients who ingested ≥40 g of immediate release paracetamol over less than eight hours within four hours of their last ingestion (odds ratio 0.12, 95% CI <0.001 to 0.91) compared with patients who did not receive activated charcoal within the four hour time window.[45] In cases of massive overdose, the window for benefit potentially extends beyond four hours if undigested drug remains in the gastrointestinal tract. No data support multi-dose activated charcoal to enhance paracetamol elimination.

Acetylcysteine

Acetylcysteine is the mainstay of treatment. Mechanisms of action differ by timing:

- Early administration (<8-10 hours): Prevents injury by replenishing glutathione stores, which detoxifies NAPQI before it binds to mitochondria. Acetylcysteine is almost universally effective in this window[46]
- Late administration (>10-24 hours): NAPQI may still be being produced. In addition, restoration of glutathione stores is essential for mitochondrial function and liver regeneration, and improves microcirculatory flow in established failure, likely improving survival even in fulminant cases.[47]

Acetylcysteine regimens

Intravenous (IV) acetylcysteine is preferred over oral routes in most settings, as oral acetylcysteine is emetogenic.[48] The traditional 3-bag regimen delivers 300 mg/kg acetylcysteine over 21 hours but is associated with higher rates of treatment delays and anaphylactoid reactions than are seen with 2-bag regimens.[49] Anaphylactoid reactions are non-allergic reactions caused by histamine release driven by the high concentration and rate of the initial acetylcysteine infusion. Clinically, anaphylactoid reactions present as flushing, urticaria, nausea, vomiting, and bronchospasm; angio-oedema and hypotension are less common. In contrast, features such as airway compromise, refractory hypotension, or cardiovascular instability (especially in patients with previous uneventful acetylcysteine exposure or known severe allergies) should raise concern for true anaphylaxis rather than a rate related anaphylactoid reaction and prompt urgent intervention.

With the traditional 3-bag regimen, severe anaphylactoid reactions occur in approximately one in three patients, typically within the first 15-30 minutes of the initial infusion (bag 1).[50] With newer 2-bag regimens, the incidence is reduced to approximately one in 20 patients (relative risk 0.18 with the SNAP regimen), consistent with the reduced rate of the initial infusion.[50] Importantly, anaphylactoid reactions do not represent a contraindication to further treatment.

Clinicians should move away from the traditional 21 hour, 3-bag regimen, and advocate for local guidelines to be updated to a 2-bag regimen. Two 2-bag regimens are now widely adopted and are described in table 1. Both have shown key improvements in care, though they have not been directly compared:

Table 1 | Comparison of common intravenous acetylcysteine regimens

Regimen	Duration	Dosing summary	Status
Traditional 3-bag	21 hours	Bag 1: 150 mg/kg (1 h) Bag 2: 50 mg/kg (4 h) Bag 3: 100 mg/kg (16 h)	Historical standard. High rate of adverse effects
12 hour modified 2-bag (SNAP)	12 hours	Bag 1: 100 mg/kg (2 h) Bag 2: 200 mg/kg (10 h)	Standard in UK. Shown to reduce adverse effects in randomised controlled trials and observational study
20 hour modified 2-bag (Australia/ New Zealand)	20 hours	Bag 1: 200 mg/kg (4 h) Bag 2: 100 mg/kg (16 h)	Standard in Australia and New Zealand. Shown to reduce adverse effects in observational study. Non-inferiority randomised controlled trial supports stopping at 12 hours for some patients

Any change to your current regimen requires system-wide management of change and should be based on specialty guidelines where available. Worked example for a 70 kg patient: Traditional 3-bag: Bag 1=10.5 g (150 mg/kg) over 1 h; Bag 2=3.5 g (50 mg/kg) over 4 h; Bag 3=7 g (100 mg/kg) over 16 h; total=21 g (300 mg/kg) over 21 h. SNAP 12 h 2-bag: Bag 1=7 g (100 mg/kg) over 2 h; Bag 2=14 g (200 mg/kg) over 10 h; total=21 g (300 mg/kg) over 12 h. Australian 20 h 2-bag: Bag 1=14 g (200 mg/kg) over 4 h; Bag 2=7 g (100 mg/kg) over 16 h; total=21 g (300 mg/kg) over 20 h.

- The 12 hour modified 2-bag regimen (SNAP) delivers 300 mg/kg over 12 hours (bag 1: 100 mg/kg over 2 hours; bag 2: 200 mg/kg over 10 hours). A randomised controlled trial involving 217 adults with acute paracetamol toxicity across 3 UK based hospitals showed that this regimen statistically significantly reduced anaphylactoid reactions and vomiting compared with the 21 hour regimen; severe anaphylactoid reactions fell from 31.0% to 5.6% (OR 0.23, 97.5% CI 0.12 to 0.43) and vomiting from 65% to 36% (adjusted OR 0.26, 97.5% CI 0.13 to 0.52).[50] Subsequent large audits and secondary analyses have validated the findings and confirmed that patients of all ages benefit from the improved side effect profile.[4,51] This is standard care in the UK, and has been externally validated as reducing anaphylactoid reactions and reducing time to hospital discharge.[52,53]
- The 20 hour 2-bag regimen (Australia/New Zealand) delivers 300 mg/kg over 20 hours (bag 1: 200 mg/kg over 4 hours; bag 2: 100 mg/kg over 16 hours). An observational study found that this regimen significantly reduced anaphylactoid reactions and vomiting compared with the 3-bag regimen; anaphylactoid reactions fell from 31% to 19% (P<0.001).[54] Two non-inferiority trials have shown that in lower risk patient cohorts (defined as "patients presenting within eight hours of ingestions <30 g," or "patients demonstrating ALT<40 U/L at presentation, and ALT<40 U/L *and* paracetamol concentration <20 mg/L after 12 hours of acetylcysteine"), the regimen is non-inferior even when shortened to 12 hours (250 mg/kg).[55,56] This regimen is standard care in Australia and New Zealand.[27]

Where no local protocol exists, we suggest clinicians use one of these 2-bag protocols, as the literature supports fewer adverse events. Pragmatic adaptations may be required if local laboratory services cannot quantify paracetamol concentrations.

Acetylcysteine stopping criteria

Typical stopping criteria depend on assessment of hepatocellular injury trajectory, liver synthetic function, and whether there is unmetabolised paracetamol remaining. Specific scenarios and international guidelines differ in their stopping criteria, but these are beyond the scope of this article.[57] As a general example, treatment could be stopped if, following 12 hours of acetylcysteine infusion:

- ALT is not above the upper limit of normal, and has not doubled or more from admission (even within normal range), *and*
- INR ≤ 1.3 (indicating normal synthetic function), *and*
- Paracetamol concentration ≤10 mg/L.

In cases of ALT rise, treatment should be continued until the ALT has begun to fall, and any INR rise has recovered to ≤1.3.[23] If locally accepted stopping criteria are not met, further acetylcysteine is typically prescribed at the rate of the second bag.

Fluid management

Acetylcysteine protocols diluted in 5% dextrose have been associated with dilutional hyponatraemia leading to seizures and coma in case reports. This is primarily a concern in adult and paediatric patients with bodyweight under 40-50 kg, in whom (if dextrose is used) the volume of diluent represents a proportionally large free water load. Observational studies and clinical protocols therefore recommend diluents that contain sodium (eg, saline based solutions) to mitigate this risk.[58] In our view, the severity of potential adverse outcomes justifies precautionary use of crystalloid diluents (eg, 0.9% saline) and/or reduced volume infusions pre-calculated according to local protocol to reduce dosing errors.[23]

Acute liver failure

Patients who develop severe acute liver injury or failure require urgent specialist care. In paracetamol toxicity, the high risk criteria are distinct: a high INR (>3.0, indicating impaired liver function) or ALT (>1000 U/L, indicating a significant degree of hepatocellular necrosis) indicate severe injury even before encephalopathy develops.[8]

Management

- Discuss all patients with severe liver injury (eg, INR > 3.0, rising creatinine, acidosis, or any sign of encephalopathy) with a specialist liver transplant unit early in the clinical course.[8] The window for safe transfer can be narrow, owing to rapid deterioration. The King's College Criteria (KCC) and modified KCC have historically been widely used to identify patients who may require referral to transplant services, incorporating pH, INR, creatinine, bilirubin, and encephalopathy grade. While these tools remain embedded in many guidelines and educational curricula in the UK, they are imperfect (sensitivity 58%, (95% CI, 51% to 65%), specificity 89% (95% CI, 85% to 93%)), and they have been largely superseded by dynamic reassessment using evolving laboratory parameters and clinical trajectory. Seek specialist input early rather than waiting for acute liver failure criteria to be met.[8,59]
- Continue acetylcysteine at the rate of your chosen regimen's last bag.
- Monitor strictly for hypoglycaemia (failure of gluconeogenesis), metabolic acidosis, and renal dysfunction. Treat hypoglycaemia with IV dextrose.[60]

A major gap in management is an inability to accurately predict the trajectory of established liver injury (distinguishing those who will recover spontaneously from those who may require transplantation) and a lack of therapies other than orthotopic liver transplantation in established liver failure.[34,37]

Supporting improved patient centred care

Most paracetamol overdoses occur in the context of self-harm. This is a dual medical and psychiatric emergency. Conduct mental health assessments concurrently with medical treatment; do not delay mental health evaluation and intervention for "medical clearance."[61] Validating patients' distress is a priority.[62] Patients at medium or high risk of suicide should be observed closely in the emergency department.[63]

Newer 2-bag regimens reduce anaphylactoid reaction rates compared with the traditional 3-bag regimen, owing to the slower initial infusion rate.[4,50] If a reaction occurs, do not stop treatment permanently. Pause the infusion, administer an antihistamine and anti-emetic, and restart at a slower rate once symptoms settle. This approach prioritises liver protection while managing patient comfort.

Case conclusion

You validate the patient's distress regarding her previous reaction. You reassure her that she had experienced a common, non-allergic anaphylactoid reaction, likely caused by the speed of the previous infusion. You explain that your hospital now uses the 12 hour SNAP regimen, which uses a slower initial infusion rate and has been shown to reduce the incidence of anaphylactoid reactions from

31.0% to 5.6%.[50] You advise that, given her high paracetamol level and current evidence, this is the safest way to prevent liver failure. You also arrange for a concurrent mental health assessment to occur during her treatment.

Education into practice

- Does your institution still use the traditional 3-bag, 21 hour acetylcysteine regimen in paracetamol overdose? What steps would be required to update this?
- How could your local treatment guidelines better address patient experience? Reflecting on the last patient you treated for paracetamol overdose: was mental health assessment initiated concurrently with medical treatment, or was it delayed pending "medical clearance"? If delayed, what barriers prevented concurrent assessment?

How patients were involved in the creation of this article

AR is a co-author and member of the Sheffield Emergency Care Forum. Her involvement ensured the article maintained a patient centred focus, particularly regarding side effects reduction, validating patient distress, and the necessity of co-delivered mental health care.

How this article was created

The scope of the article was defined after suggestion of relevant domains by the authors. Authors undertook focused literature searches to identify relevant new evidence from within the past 10 years. Where key components of clinical care lacked new evidence, foundational articles were identified and referenced. Additional clinical guidelines known to the authors were included when relevant. A key focus of the article was to signpost interested readers to additional information for detailed discussion.

If you're struggling, you're not alone. In the UK and Ireland Samaritans can be contacted on tel 116 123 or at https://www.samaritans.org/how-we-can-help/contact-samaritan for an online chat service. In the US, the National Suicide Prevention Lifeline is 1-800-273-8255. In Australia, the crisis support service Lifeline is 13 11 14. Other international helplines can be found at www.befrienders.org.

Contributorship and guarantor: CH and JD conceived the article and are guarantors. All authors (CH, IG, MG, AR, JD) made substantial contributions to the design of the work, drafting and critical revision, final approval of the version to be published, and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. AR was the author group's patient and public representative, and participated as an author.

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